

NBA_EDA

September 14, 2025

```
[2]: import pandas as pd
import numpy as np
data = pd.read_csv("NBA.csv")
data.info()
#calculate the missing values
data.isna().sum().sort_values()
```

<class 'pandas.core.frame.DataFrame'>

RangeIndex: 219524 entries, 0 to 219523

Data columns (total 36 columns):

#	Column	Non-Null Count	Dtype
0	GRID_TYPE	219524 non-null	object
1	GAME_ID	219524 non-null	int64
2	GAME_EVENT_ID	219524 non-null	int64
3	PLAYER_ID	219524 non-null	int64
4	PLAYER_NAME	219524 non-null	object
5	TEAM_ID	219524 non-null	int64
6	TEAM_NAME	219524 non-null	object
7	PERIOD	219524 non-null	int64
8	MINUTES_REMAINING	219524 non-null	int64
9	SECONDS_REMAINING	219524 non-null	int64
10	ACTION_TYPE	219524 non-null	object
11	SHOT_TYPE	219524 non-null	object
12	SHOT_ZONE_BASIC	219524 non-null	object
13	SHOT_ZONE_AREA	219524 non-null	object
14	SHOT_ZONE_RANGE	219524 non-null	object
15	SHOT_DISTANCE	219524 non-null	int64
16	LOC_X	219524 non-null	int64
17	LOC_Y	219524 non-null	int64
18	GAME_DATE	219524 non-null	object
19	HTM	219524 non-null	object
20	VTM	219524 non-null	object
21	clock	219524 non-null	object
22	periodType	219524 non-null	object
23	actionType	219524 non-null	object
24	x	219524 non-null	float64
25	y	219524 non-null	float64

```

26  scoreHome          219524 non-null  int64
27  scoreAway          219524 non-null  int64
28  side               219524 non-null  object
29  shotResult         219524 non-null  int64
30  pointsTotal        219524 non-null  int64
31  assistPlayerNameInitial 219524 non-null  object
32  assistPersonId     219524 non-null  int64
33  assistTotal        219524 non-null  int64
34  SCORE_MARGIN       219524 non-null  int64
35  MARGIN_ACTION      219524 non-null  int64
dtypes: float64(2), int64(18), object(16)
memory usage: 60.3+ MB

```

```

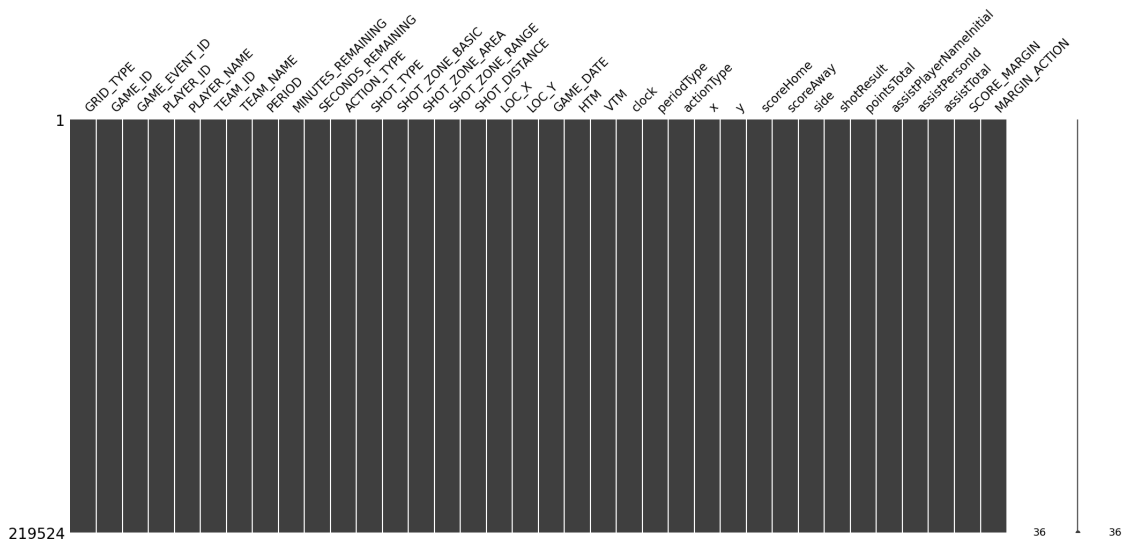
[2]: GRID_TYPE          0
    GAME_ID             0
    GAME_EVENT_ID       0
    PLAYER_ID           0
    PLAYER_NAME          0
    TEAM_ID             0
    TEAM_NAME           0
    PERIOD              0
    MINUTES_REMAINING    0
    SECONDS_REMAINING    0
    ACTION_TYPE         0
    SHOT_TYPE           0
    SHOT_ZONE_BASIC      0
    SHOT_ZONE_AREA       0
    SHOT_ZONE_RANGE      0
    SHOT_DISTANCE        0
    LOC_X               0
    LOC_Y               0
    GAME_DATE           0
    HTM                 0
    VTM                 0
    clock               0
    periodType          0
    actionType          0
    x                   0
    y                   0
    scoreHome           0
    scoreAway           0
    side                0
    shotResult          0
    pointsTotal         0
    assistPlayerNameInitial 0
    assistPersonId      0
    assistTotal         0

```

```
SCORE_MARGIN          0
MARGIN_ACTION         0
dtype: int64
```

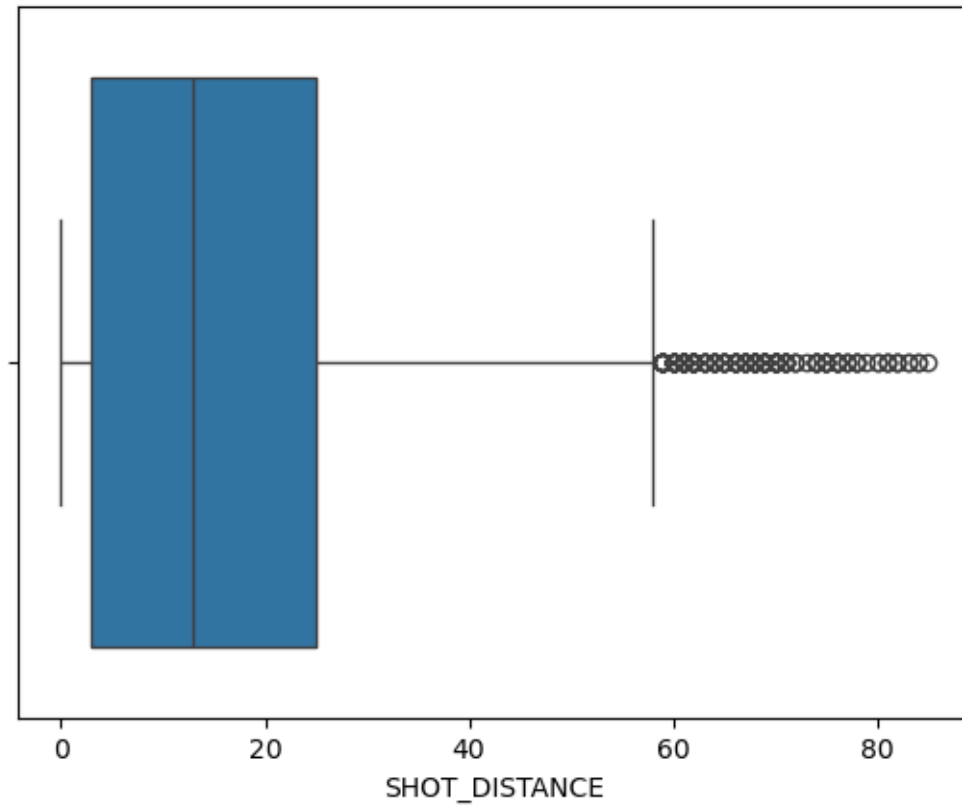
```
[3]: import missingno as msno
      msno.matrix(data)
```

```
[3]: <Axes: >
```



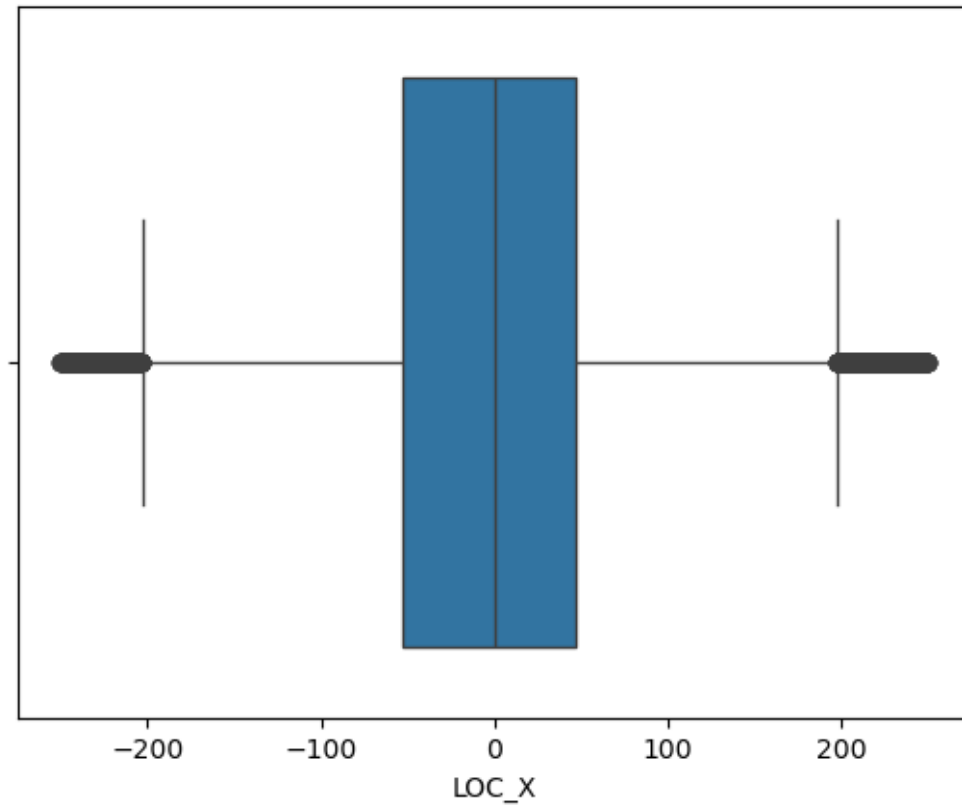
```
[4]: import seaborn as sns
      import matplotlib.pyplot as plt
      #boxplot
      sns.boxplot(x = data["SHOT_DISTANCE"])
      """Most shot_distance are in 0-30feet.
         The median is around 10 to 15.
         SHOT_DISTANCE over 60 are relatively
         rare and are treated as outliers."""
```

```
[4]: 'Most shot_distance are in 0-30feet.\n    The median is around 10 to 15.\n    SHOT_DISTANCE over 60 are relatively\n    rare and are treated as outliers.'
```



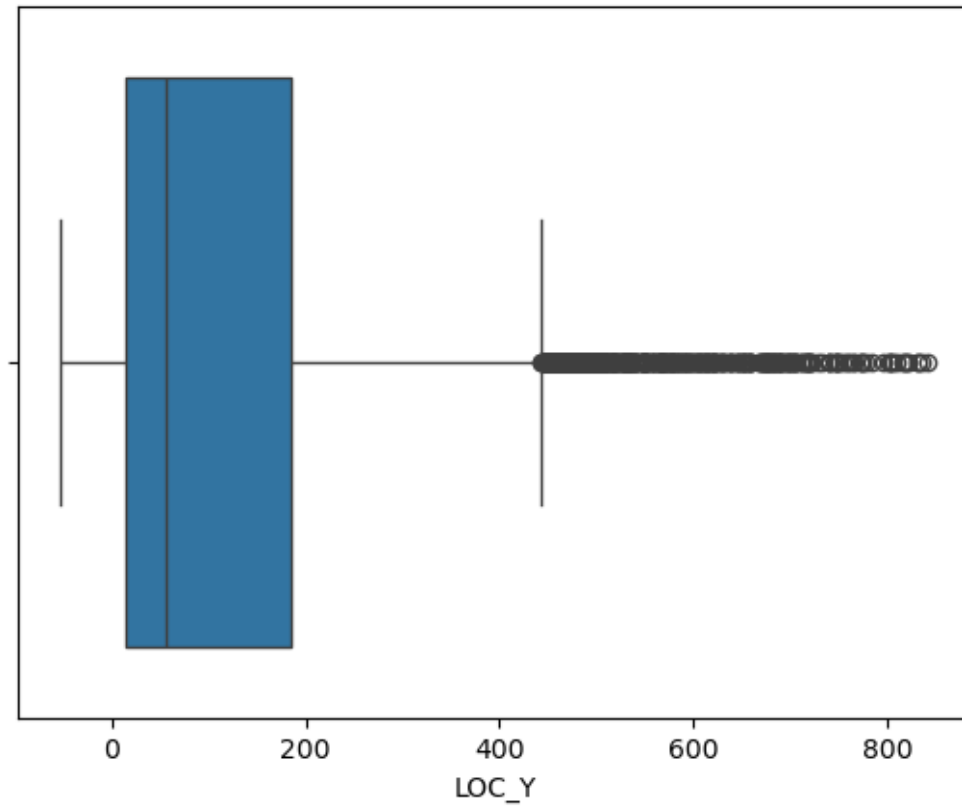
```
[5]: sns.boxplot(x = data["LOC_X"])
      """Most shot horizontal positions are
      concentrated within 50feet to the
      left and right of the basket.
      The extreme points(< -200 or > 200)
      are rare outliers, which may represent
      shots from the edges of court."""
```

```
[5]: 'Most shot horizontal positions are\n    concentrated within 50feet to the \n
left and right of the basket.\n    The extreme points(< -200 or > 200)\n    are
rare outliers, which may represent\n    shots from the edges of court.'
```



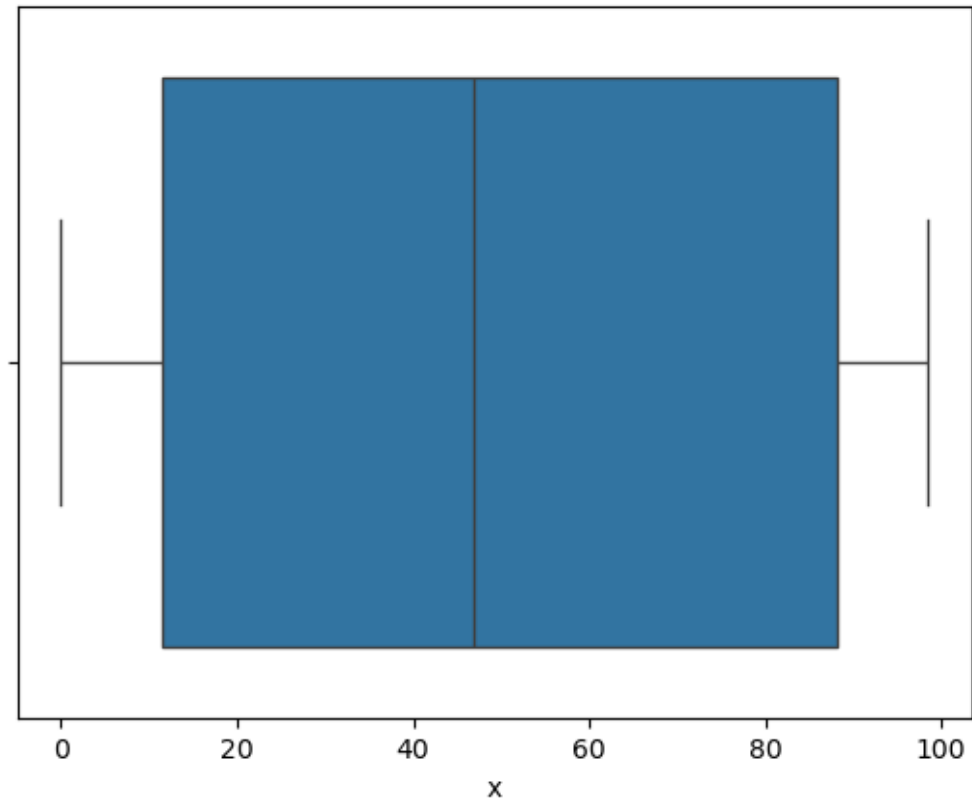
```
[6]: sns.boxplot(x = data["LOC_Y"])
      """Most shots are concentrated within the
      the 0-400 range(under the basker, mid-
      range, and beyond the three-point line.)
      The length of an NBA half-court is about
      470feet."""
```

```
[6]: 'Most shots are concentrated within the\n    the 0-400 range(under the basker,
mid-\n    range, and beyond the three-point line.)\n    The length of an NBA half-
court is about\n    470feet.'
```



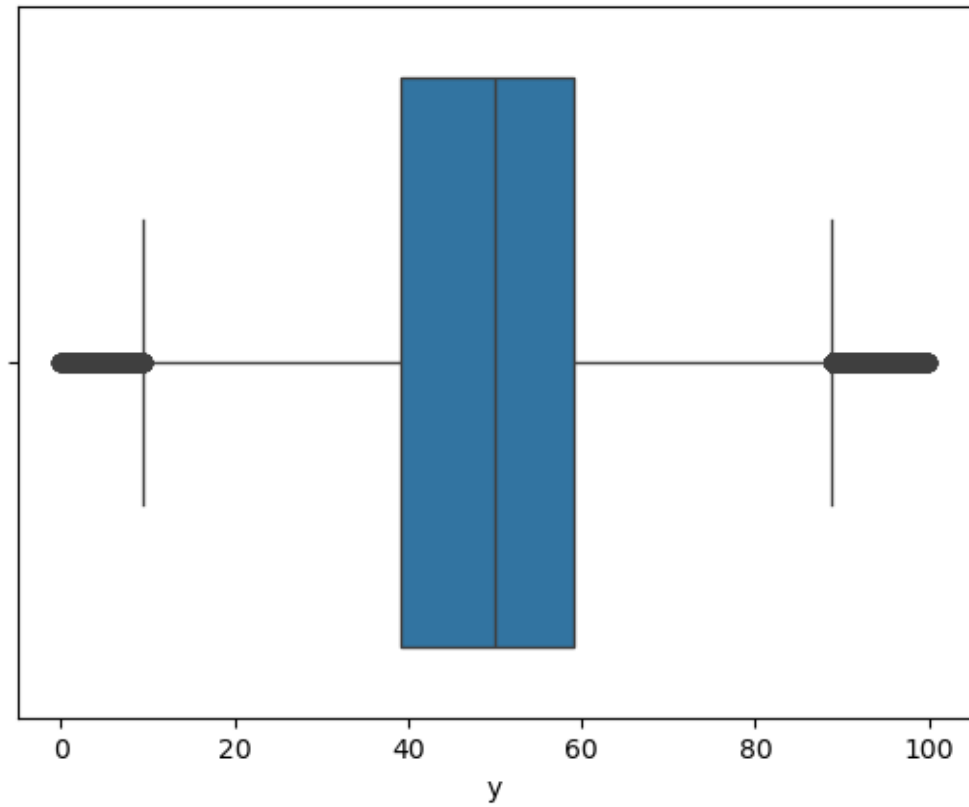
```
[7]: sns.boxplot(x = data["x"])
```

```
[7]: <Axes: xlabel='x'>
```



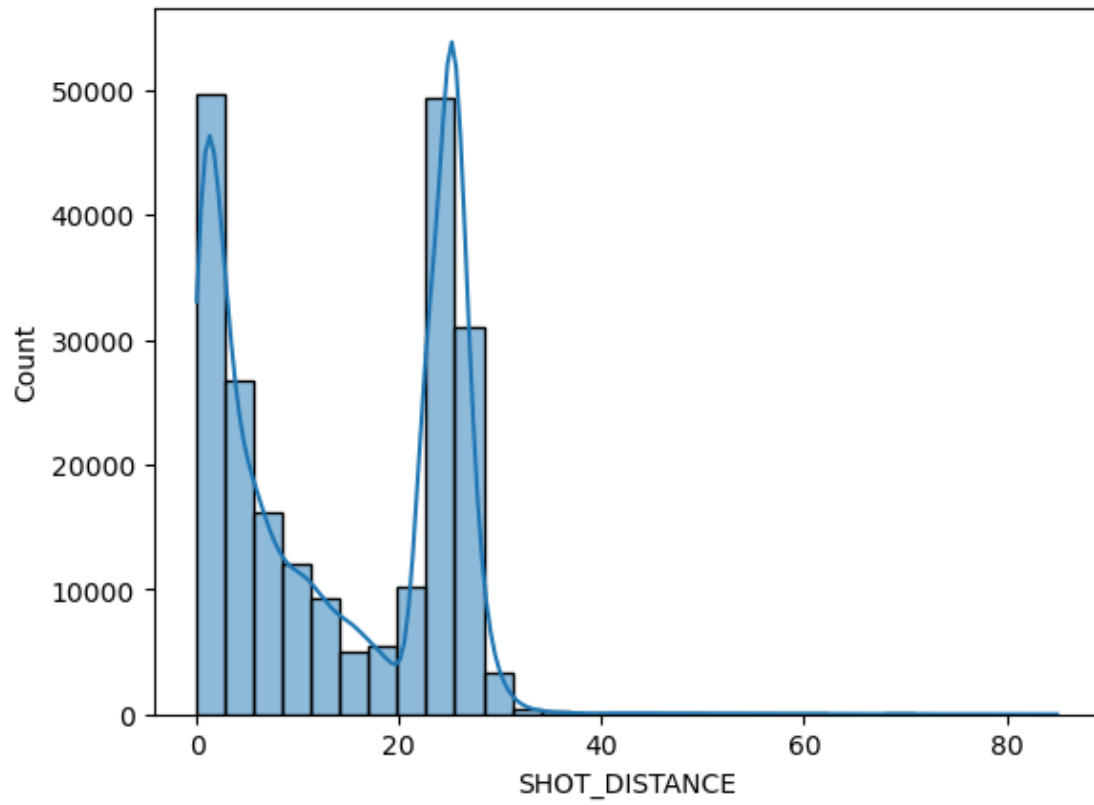
```
[8]: sns.boxplot(x = data["y"])
```

```
[8]: <Axes: xlabel='y'>
```



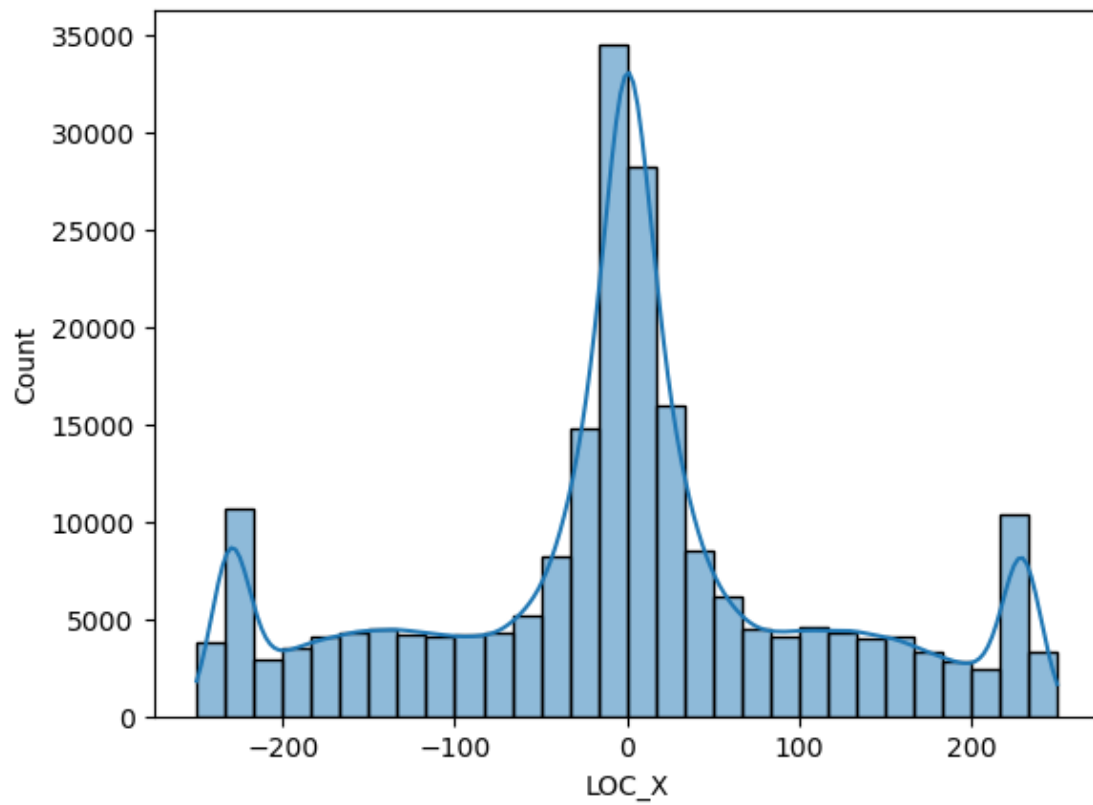
```
[10]: #data visualization
sns.histplot(data['SHOT_DISTANCE'], bins = 30, kde = True)
```

```
[10]: <Axes: xlabel='SHOT_DISTANCE', ylabel='Count'>
```

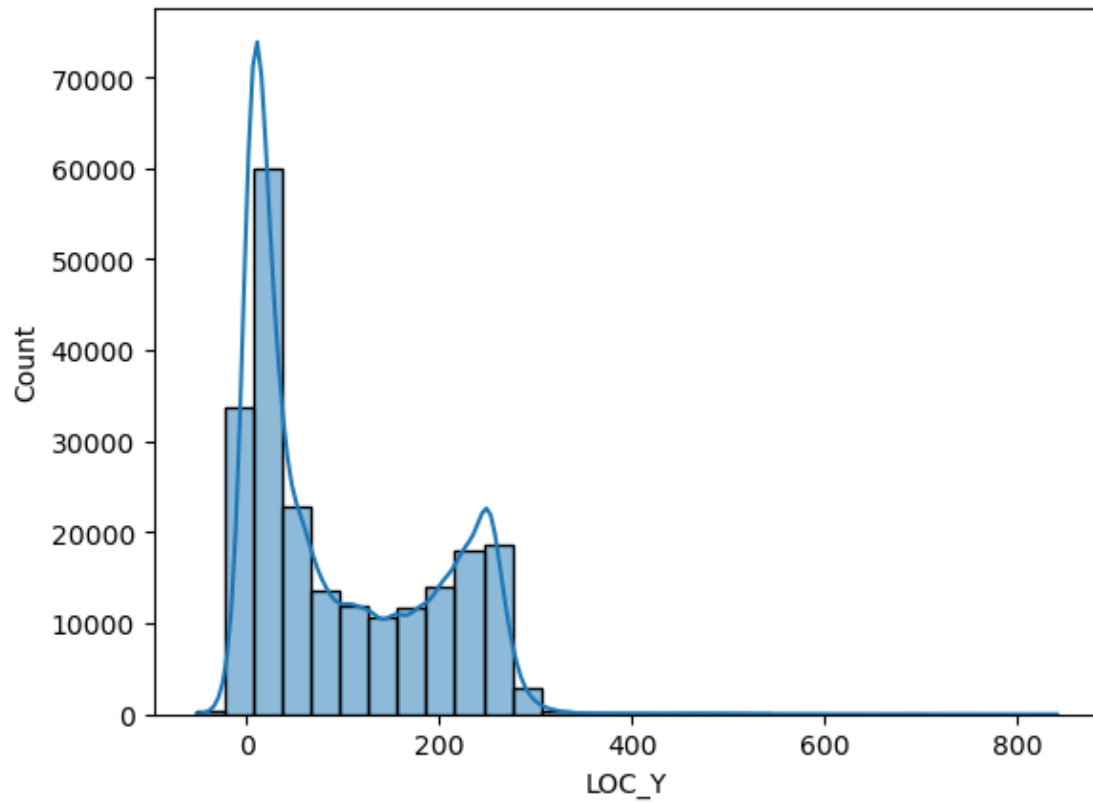
```
[11]: sns.histplot(data['LOC_X'], bins = 30, kde = True)
```

```
[11]: <Axes: xlabel='LOC_X', ylabel='Count'>
```

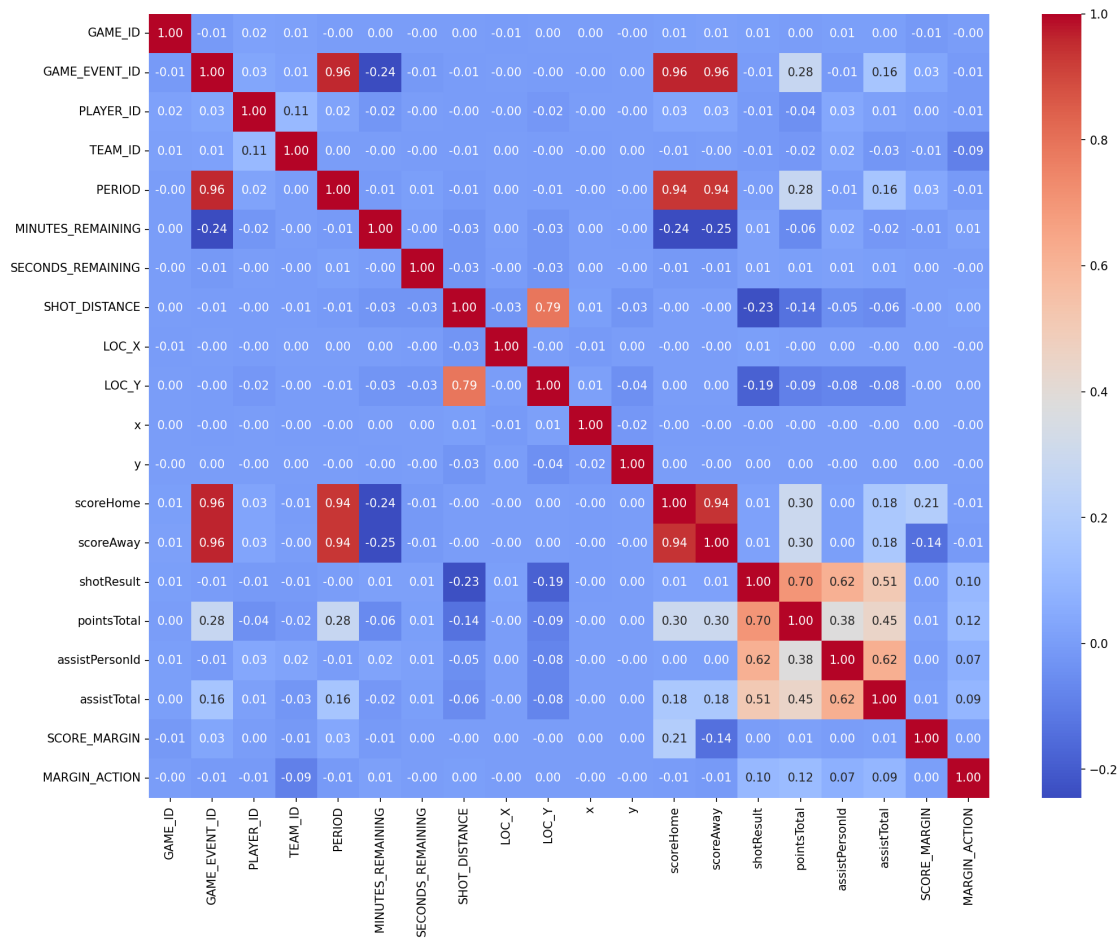


```
[12]: sns.histplot(data['LOC_Y'], bins = 30, kde = True)
```

```
[12]: <Axes: xlabel='LOC_Y', ylabel='Count'>
```



```
[13]: #Correlation Coefficient & Heatmap
corr = data.corr(numeric_only=True)
plt.figure(figsize=(16, 12), dpi=150) # dpi=
sns.heatmap(corr, annot=True, fmt=".2f", cmap="coolwarm")
plt.show()
```



[]: